BACHELOR IN SOFTWARE ENGINEERING (HONOURS)

FINAL EXAMINATION
JANUARY 2021

Course : EC3307 (Object and Component Technology) Time: 1.30pm – 4.30pm
(3 hours)
Lecturer : Rajesvary Rajoo Date : 5 May 2021

Instructions:

Answer **ALL** questions.

This examination paper is confidential. The questions must be answered individually. Students are NOT PERMITTED to discuss or consult with other students or individuals.

Using Google is not allowed at all. Thus, the answer from the Internet will be considered plagiarism. Plagiarism is an offence. University guidelines on plagiarism will apply.

All exams submitted are final. Students will NOT BE PERMITTED to submit any additional work or alternative version, even if time is remaining. Only the initial submission will be forwarded for grading.

Your answer MUST be submitted within the stipulated time. Failure to submit your answers within the deadline given may result in the award of zero marks. You will be given an additional 20 minutes at the end of the specified exam duration. This extra time is for you to submit and upload your completed exam. It's not intended as extra working time. If you experience technical difficulties, you can use this time at your own discretion, but you must leave sufficient time to submit and upload your completed exam.

Answer Format:

i. Do not put your name on any materials related to the exam. Use only your Student ID Number for identification.
ii. All answers must be handwritten, scanned (using camscanner) and converted to PDF file.
iii. Save your answers in the following format: **STUDENT ID_COURSE CODE_COURSE TITLE**

Honor Pledge for Exams
"I affirm that I have not given or received any unauthorised help on this exam, and that all work is my own."

Name and Signature: ___________

This question paper consists of **3** pages.
(excluding front cover)

1. Consider the below specification of an abstract data type (ADT) Queue with some of its methods:

```
public interface Queue <E> {
/* A Queue <E> object is a list whose elements are of type E.*/

        public int count ();
        /*      Return the size of the queue. */
        public E peek ();
        /*      Returns the head element of the queue */
        public void add (E x);
        /*    Adds the specified element at the end of queue.
              Increases the count */
        public E remove ();
        /*     Returns and removes the head element of the queue.
               Decreases the count  */
        public boolean isEmpty();
        /*     Returns true if empty, false if not */
        public Iterator<E> iterator ();
        /*     Return an iterator that will visit the elements of this list from left to right*/
        }

}
```

a. Write the pre-condition and post-condition for method remove () for the Queue ADT. **(5 MARKS)**

b. Explain the relationship between the precondition and post-condition of Hoare logic and the contract of Meyer's 'design by contract' for method remove for the Queue ADT. **(5 MARKS)**

c. Classify the methods of Queue ADT into queries and operations. Explain your choices. **(4 MARKS)**

d. Explain how a client (user) of the Queue ADT could ensure that a call of method remove conforms to the 'contract' implied by the specification of Queue ADT. **(5 MARKS)**

e. Explain what it meant by "pre-condition is false" and "precondition is true". **(6 MARKS)**

2. a. Explain by using a suitable example how Design Patterns help software developers. **(7 MARKS)**

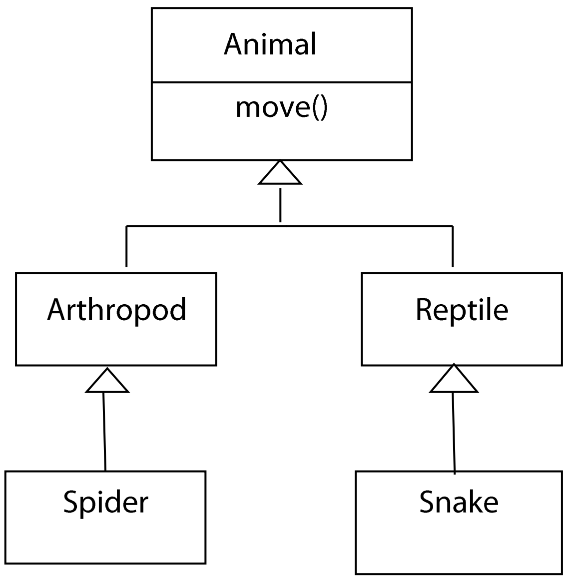
b. Examine the following scenario:

"Suppose that a military complex has software that includes a class called Personnel, objects of which represent the people who work there. Suppose further that you were asked to change the software so that the people are tagged with sensors that reveal their location, as represented by objects of some class Location. The old methods of Personnel, however, must be accessible just as before."

i. Explain what Design Pattern you would use for this modification and why. **(4 MARKS)**

ii. Design the solution in English, Java or UML. **(5 MARKS)**

c. Consider the following simple class hierarchy:



i. Reify the method **"move"** to the pattern Strategy. The different types of move are crawl, slither and spit venom . **(5 MARKS)**

ii. Describe the design principle that is being used in c(i). Explain the advantage of using it. **(4 MARKS)**

3. a. Explain an advantage of using generic parameter (type parameter) in programming. Illustrate your answer using an example. **(7 MARKS)**

   b. Consider the following code. Find the error and explain what will happen if you try to compile and execute the code. **(4 MARKS)**

```
Stack<Object> stack = new Stack<Object>( );
stack.push("hello");
String s = stack.pop( );
```

   c. Explain the difference between an int and an Integer in Java. **(4 MARKS)**

   d. Write generic method, using a generic type parameter <T>, that replaces every occurrence in a ArrayList<T> of a specified item with a specified replacement item. The list and the two items are parameters to the method. Both items are of type T. Take into consideration the fact that the item that is being replaced might be null. For non-null item, use equals() to do the comparison. **(10 MARKS)**

4. a. Contrast the classic Java approach of the Java Virtual Machine with the .NET way of using an intermediate language. **(4 MARKS)**

   b. With Microsoft's .NET it is possible to compose a program from components written in several different programming languages. Illustrate how this is made possible by using an example. Give any **TWO (2)** advantages of this. **(8 MARKS)**

   c. Consider the following statement: **(8 MARKS)**

   "Limited reintroduction of the goto in programming is not considered harmful if used wisely."

   Do you agree with this statement? Give a simple code example to illustrate your answer. Otherwise, give a counter argument with an example code.

   d. Explain the difference between stack memory and heap memory with reference to deciding where to store objects. **(5 MARKS)**

-END OF QUESTION PAPER-